SI 313 - Intro to Quantitative Methods - WN26
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Big vs. Huge Cities: Can 311 data predict socioeconomic characteristics in Pittsburgh?

1. Introduction

Put simply, 311 data are service requests made by city residents to local governments, usually through a dedicated phone number (311) or online platforms (e.g. SeeClickFix). This data helps cities track issues like noise complaints, whether street repairs are needed, and any other community concerns, which helps allow for better service around the city.

311 data can also be used for more than city planning and tracking service requests. In "Structure of 311 service requests as a signature of urban location,"[1] Wang, et al. ask if 311 service request patterns can predict socioeconomic characteristics of neighborhoods in large urban cities, and specifically look at whether neighborhoods with similar demographics (like income, race, and education) produce similar 311 complaint patterns. The paper also aimed to learn if these requests could be used to create an accurate model of socioeconomic performance of urban neighborhoods in general as a tool for "urban stakeholders to quantify the impacts of their interventions" (1).

In 2017, Wang, et al. did a large-scale analysis of 311 service request data from New York, Chicago, and Boston spanning from 2012-2015. They used machine learning models to group census tracts by relative frequencies of complaint types, and created "complaint signatures" from this grouping. They then compared the data against large demographic data from the American Community Survey (ACS) census. They found neighborhoods in these cities have distinct signatures–wealthier areas typically report things like noise complaints, and lower-income areas have higher frequencies of blocked driveway and heating issue complaints–and that 311 data patterns could very accurately predict factors like median income, education level, and the racial demographics of an area as they're represented in ACS census data. Past research like this shows that 311 data can be effective in extremely densely populated, metropolitan areas like New York (population: ~8.5m), but can 311 service request patterns in smaller cities like Pittsburgh (population: ~300k) predict socioeconomic characteristics of its neighborhoods in a similar way the data does in much larger cities?

This paper iterates on the methods used by Wang, et al., but applies the concepts to 311 data in the neighborhoods of Pittsburgh, Pennsylvania. It compares Pittsburgh 311 requests with ACS census data to see if correlations appear based on socioeconomic characteristics of Pittsburgh neighborhoods. 311 and other similar request data are valuable data sources; they're

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[1] Wang, Lingjing, Cheng Qian, Philipp Kats, Constantine Kontokosta, and Stanislav Sobolevsky. 2017. "Structure of 311 Service Requests as a Signature of Urban Location." PLoS ONE 12 (10): e0186314. https://doi.org/10.1371/journal.pone.0186314.

public, huge (place-dependent, but generally speaking), and granular. Using 311 and existing census data to predict the socioeconomic characteristics of neighborhoods in cities like Pittsburgh has the potential to be a cheaper and quicker supplement to slow and expensive census data. Verifying a relationship between 311 data and socioeconomic characteristics might help urban planners and researchers looking to use this data to inform business decisions and research. This research finds that there does not appear to be a significant relationship between median income levels and 311 request categories in Pittsburgh, and therefore that 311 data cannot be used to create "complaint signatures" in smaller or medium sized cities like Pittsburgh as proposed for large cities like New York City in the Wang et al. paper.

2. Methods

2.1 Research Design and Data Source

This paper is an extension and replication of the 2017 study by Wang et al., “Structure of 311 service requests as a signature of urban location.” The original study focused on extremely densely populated metropolitan areas like New York. This research scales down the methods to Pittsburgh to determine if neighborhoods in Pittsburgh with similar demographics (specifically median income) produce distinct 311 complaint “signatures,” as well as to determine if this can be generalized to other areas around Pittsburgh’s size.

This research uses public 311 service request data from Pittsburgh and ACS census data with the census tracts of neighborhoods in Pittsburgh for 2024. The 311 data is found on the Western Pennsylvania Regional Data Center’s (WPRDC) public 311 data archive on wprdc.org.[2] The census data is from the ACS 2024 1-year estimate for the census tracts within Pittsburgh, PA. It’s found on the United States Census Bureau's online data table on census.gov.[3] Additionally, since the WPRDC data does not provide census tract details in the same way as the ACS data, the WPRDC data was mapped to a ‘block to neighborhood index’ for 2020 Census geographies and neighborhoods, created by Chris Briem. It’s also available from the WPRDC.[4]

2.2 Measures / Operationalization

The independent variable, socioeconomic status, was operationalized by using the ACS Median Household Income data. Neighborhoods were divided into three “income groups” (Low,

[2] “311 Data Archive - 311 Data - CKAN.” n.d. https://data.wprdc.org/dataset/311-data/resource/29462525-62a6-45bf-9b5e-ad2e1c06348d.

[3] U.S. Census Bureau. n.d. “Explore Census Data.” https://data.census.gov/table?q=DP03&g=160XX00US4261000,4261000$1400000&d=ACS+5-Year+Estimates+Data+Profiles.

[4] “2020 Census Redistricting Data Extracts (PL 94-171) - Block Index City of Pittsburgh Neighborhoods - CKAN.” n.d. https://data.wprdc.org/dataset/2020-census-redistricting-data-extracts/resource/6b09ea3e-7d34-4665-ad0b-798a0efadc29.

Medium, High Income) based on quartiles or “bins,” with the Low Income group representing households with median incomes from $0 to $68,708 (minimum to Q2), Medium Income group with incomes from $68,708 to $83,500 (Q2 to Q3), and High Income group with incomes from $83,500 and up (Q3 and up, max of $222,778).

The dependent variable, 311 request type, was operationalized as the relative frequency of 311 request category. The dataset originally represented 247 unique complaint types, and these were consolidated into 24 broader “categories” using Google Gemini, a sample of which are shown in table 1a (23 named categories, plus an “other” category).

Table 1a: Complaint type mapping (sample)
Full table available in appendix

Category	Mapped Complaint Types
Refuse & Sanitation	Missed Refuse Pick Up, Missed Recycling Pick Up, Bulk Pick Up, Special Pick Up, Courtesy Pick Up, Illegal Dumping, Dumping, Private Property, Early Set Out, Late Set Out, Litter, Litter, Public Property, Weeds/Debris, Recycling Violation, Refuse Violations, Blue Bin Containers, Dumpster, Dumpster (on Street), Commercial Refuse/Dumpsters, Litter Can, Litter Can, Public, Dead Animal, Animal Waste, Rodent control
Street & Infrastructure	Potholes, Paving Concern/Problem, Paving Request, Brick or Block Repair, Concrete Slab Repair, Street Cleaning/Sweeping, Landslide, Sinkhole, Bridge Maintenance, Curb Cuts, Curb/Request for Asphalt Windrow, Manhole Cover, Manhole Covers, PWSA, Utility Cut - Other, Utility Cut - PWSA, City Cuts Concern, Sidewalk/Curb/ADA Ramp Maintenance, Broken Sidewalk, Repair City Steps
Transportation & Parking	Illegal Parking, Abandoned Vehicle (parked on street), Towed Vehicle, Junk Vehicles (Private Property), Speeding, Traffic Enforcement, Traffic Calming, Loading Zone, Request, Permit Parking (Request for RPP zone), Parking Authority, Ticket, Pgh Parking Authority, Ticket, Police-issued, SPIN (Stand Up) Scooters, Boat/Trailer on Street

2.3 Analytical Techniques / Kruskall-Wallis Testing

The main analytical technique used was calculating and comparing the relative frequency of each complaint category within each income group. A Kruskall-Wallis test was run to determine if there are statistically significant differences between these independent and ordinal income groups as it relates to the average frequency of request categories. This research was initially run using a Chi-square test, but it was giving skewed results that didn’t make sense. This was happening because Chi-square compares total observed vs expected counts, not relative frequencies, and this was skewing the data because there are far more low-income data points than there were for any other income group (~12,000 vs 6,000 and 5,000 for low, medium, and high respectively). Of course this data wouldn't work with a Chi-square test; one of the groups is represented more than twice as much as the other two groups, because there are simply more people with a low income than a medium or high income in Pittsburgh. The nature of this research led me to split up the groups by income level, so there's no guarantee of an even number in each group.

Therefore, a Kruskall-Wallis test was chosen. An ANOVA test would be appropriate to test for the difference in complaint frequencies, but the data wasn’t normally distributed.

However, the Kruskall-Wallis fit perfectly, as it's used to determine if there are significant differences between two or more independent groups on a dependent variable that is ordinal or continuous (i.e. the low, medium, and high income groups) without assuming a normal distribution like ANOVA requires. This ended up working well; instead of using all 25,000 or so complaint entries from 2024 into one table with the Chi-Square test, the Kruskall-Wallis test looks at the data neighborhood by neighborhood. The test asks if the average relative frequency of a certain request category differs based on whether the neighborhood is low, medium, or high income.

The key question with this test is whether there is a statistically significant difference between income groups when it comes to how each group submits service requests, and whether "complaint signatures" can be created based on these patterns. Therefore, the research approach requires an ANOVA or Kruskall-Wallis test.

2.4 Comparison to Wang et al. Methods

While this paper is heavily inspired and guided by the methods used by Wang et al. in their 2017 study, there is an obvious difference in methodology as previously discussed. While Wang et al. used many years of 311 data points from some of the largest U.S. cities, this research scales down the scope to focus on one year (2024) in a smaller city (Pittsburgh). The purpose of this change is to determine if the findings from Wang et al.'s 2017 study can still hold in other contexts; to see if complaint signatures can be effectively generated using 311 data in smaller areas, not just in huge cities.

Wang et al. used the highly granular data available from these locations to build their complaint signatures. This wasn't as effective for a smaller city like Pittsburgh. This research required the 247 unique complaint types in the sample to be grouped into 24 unique request "categories." Additionally, the Wang et al. study looked at multiple socioeconomic characteristics, whereas this research only focuses on median income from ACS data, and separated data into income "bins": low, medium, and high income.

The original paper used in-depth machine learning algorithms to group census tracts by complaint frequencies to predict socioeconomic characteristics. They trained predictive models to guess socioeconomic characteristics purely based on 311 request signatures. This research doesn't use predictive models, but uses Kruskall-Wallis testing across the three independent income groups to determine if 311 requests can create these unique signatures.

3. Results

3.1 Descriptive Statistics

After cleaning, filtering, and merging the WPRDC and ACS data, some initial exploratory and descriptive statistics were run on the sample. The final sample held 25,153 valid 311 requests. To operationalize the independent variable, median income, each census tract was put into “bins”, separated by income tier: low, medium, or high income. These tier cutoffs were based on Q2 and Q3 of the median income from the ACS data, identified in table 4. The top five request categories across all of Pittsburgh can be seen in table 2, and a sample of the top five request categories per neighborhood can be seen in table 3a below:

Table 2: Top 5 Most Common Request Categories in Pittsburgh

Request Category	Total Count
Refuse & Sanitation	8144
Street & Infrastructure	2863
Transportation & Parking	1998
Safety & Lighting	1633
Traffic Signals & Signs	1245

Table 3a: Top 5 request categories per neighborhood (sample)

neighborhood	request_category	relative_frequency	percentage
Allegheny Center	Snow & Ice Removal	0.184834	18.48
Allegheny Center	Safety & Lighting	0.174171	17.42
Allegheny Center	Traffic Safety & Crashes	0.150474	15.05
Allegheny Center	Street & Infrastructure	0.116114	11.61
Allegheny Center	Active Transportation	0.071090	7.11

Additional statistics for the data focusing on median income can be found in table 4, and the top five request categories per income group can be found in table 5.

Table 4: Descriptive statistics - median income

measure	value
count	25153.000000
mean	72420.996303
std	36713.688650
min	8889.000000
25%	52099.000000
50%	68708.000000
75%	83500.000000
max	222778.000000

Table 5: Top 5 request categories per income group

income_group	request_category	relative_freq
Low Income	Refuse & Sanitation	0.386511
Low Income	Street & Infrastructure	0.102300
Low Income	Transportation & Parking	0.071969
Low Income	Safety & Lighting	0.065653
Low Income	Building & Property Standards	0.056686
Medium Income	Refuse & Sanitation	0.261577
Medium Income	Street & Infrastructure	0.110763
Medium Income	Transportation & Parking	0.079631
Medium Income	Safety & Lighting	0.063830
Medium Income	Traffic Signals & Signs	0.058041
High Income	Refuse & Sanitation	0.255222
High Income	Street & Infrastructure	0.142015
High Income	Transportation & Parking	0.095350
High Income	Safety & Lighting	0.064522
High Income	Parks & Greenery	0.061152

3.2 Visualizations and Statistical Testing

A heatmap and grouped bar chart were created from the data to visualize the differences between income groups. Figure 1 shows the heatmap of relative frequencies of request categories by income group, and figure 2 shows the grouped bar chart of the top five categories with the highest standard deviation in relative frequencies.

After making the heatmap, many categories looked very similar, so the categories with the highest standard deviation were used to give the best chance of determining whether 311 data could be used to predict socioeconomic characteristics. See table 6 below for the categories with the highest standard deviation and their standard deviation values.

Figure 1: Heatmap of Relative Frequencies of Request Categories by Income Group

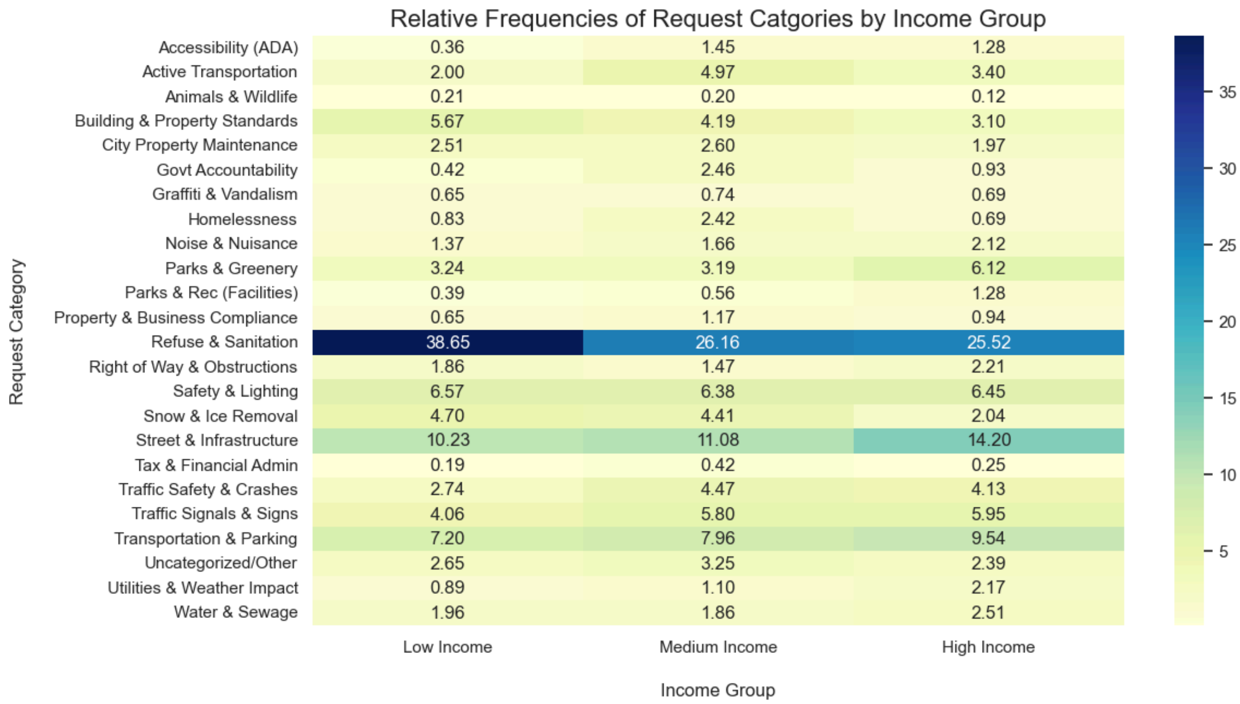


Figure 2: Grouped Bar Chart of Top 5 Request Categories with Highest Standard Deviation by Income Group

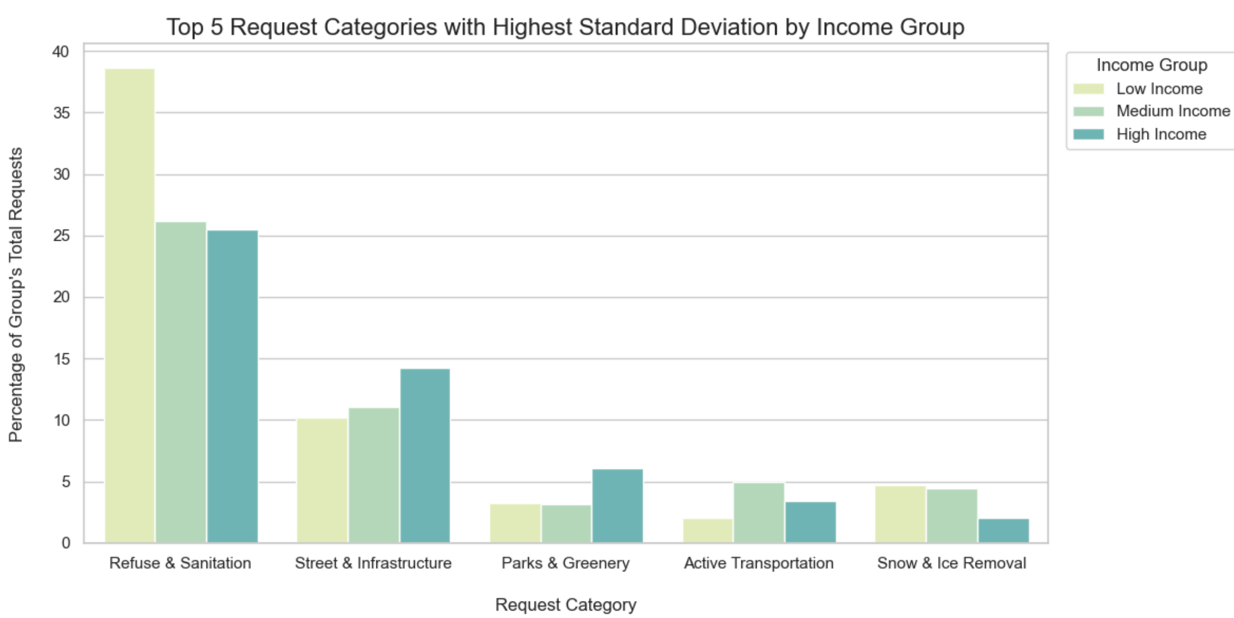


Table 6: Request Categories with Highest Standard Deviation

request_category	**std dev**
Refuse & Sanitation	7.4
Street & Infrastructure	2.09
Parks & Greenery	1.68
Active Transportation	1.49
Snow & Ice Removal	1.46

A Kruskall-Wallis test was run to determine if there was a statistically significant relationship between request category and income group. As discussed above, a Kruskall-Wallis test made the most sense because the average relative frequency of complaints was being compared across multiple independent ordinal groups. The test results can be seen in table 7:

Table 7: Kruskall-Wallis Test Results ($\alpha = 0.05$)

Request Category	**Kruskal-Wallis Statistic (χ^2)**	**P-Value**	**Conclusion**
Accessibility (ADA)	0.16	0.9238	No significant relationship
Active Transportation	4.31	0.1159	No significant relationship
Animals & Wildlife	1.07	0.5865	No significant relationship
Building & Property Standards	7.60	0.0224	**Significant relationship**
City Property Maintenance	0.06	0.9695	No significant relationship
Govt Accountability	0.11	0.9469	No significant relationship
Graffiti & Vandalism	0.04	0.9817	No significant relationship
Homelessness	7.93	0.0190	**Significant relationship**
Noise & Nuisance	4.81	0.0902	No significant relationship
Parks & Greenery	0.44	0.8028	No significant relationship
Parks & Rec (Facilities)	0.94	0.6255	No significant relationship
Property & Business Compliance	3.27	0.1945	No significant relationship
Refuse & Sanitation	14.86	0.0006	**Significant relationship**
Right of Way & Obstructions	4.35	0.1137	No significant relationship
Safety & Lighting	0.79	0.6740	No significant relationship
Snow & Ice Removal	1.84	0.3988	No significant relationship
Street & Infrastructure	0.51	0.7754	No significant relationship

Tax & Financial Admin	4.54	0.1032	No significant relationship
Traffic Safety & Crashes	4.13	0.1268	No significant relationship
Traffic Signals & Signs	2.57	0.2769	No significant relationship
Transportation & Parking	6.90	0.0317	**Significant relationship**
Uncategorized/Other	0.51	0.7767	No significant relationship
Utilities & Weather Impact	10.99	0.0041	**Significant relationship**
Water & Sewage	6.06	0.0484	**Significant relationship**

3.3 Findings / Comparison to Wang et al. Results

The statistical analysis and visualizations reveal that in general, there is no statistically significant relationship between a neighborhood's median income and the relative frequency of a given request category, with some exceptions. In other words, 311 data generally cannot be used to predict socioeconomic characteristics (like median income) in smaller cities like Pittsburgh like it can in New York, Boston, and Chicago.

When using $\alpha = 0.05$, 18 of 24 request categories revealed no significant relationship from the Kruskall-Wallis testing. With $\alpha = 0.01$, 22 of 24 request categories reveal no significant relationship.

Wang et. al found that in New York City, "complaints/requests within cluster 1 more often report noise concerns than others, cluster 2 experiences more issues relating to residential heating, cluster 3 has the highest relative complaints about blocked driveways, while cluster 4 reports concerns about street conditions" (5). So in summary, requests around noise; heat; driveways and parking; and street conditions were the types that differed the most distinctly in New York City. Interestingly, the significance revealed by the Kruskall-Wallis testing was split for these key types. As shown in table 7a, Noise & Nuisance ($\chi^2 = 4.81, p = 0.0902$) and Street & Infrastructure ($\chi^2 = 0.51, p = 0.7754$) have no significant relationship, but Transportation & Parking ($\chi^2 = 6.90, p = 0.0317$) and Water & Sewage ($\chi^2 = 6.06, p = 0.0484$) both do. Additionally, Refuse & Sanitation ($\chi^2 = 14.86, p = 0.0006$) and Homelessness ($\chi^2 = 7.93, p = 0.0190$) are categories where one may expect income level to influence request rate. It just so happens that a significant relationship was found for both categories.

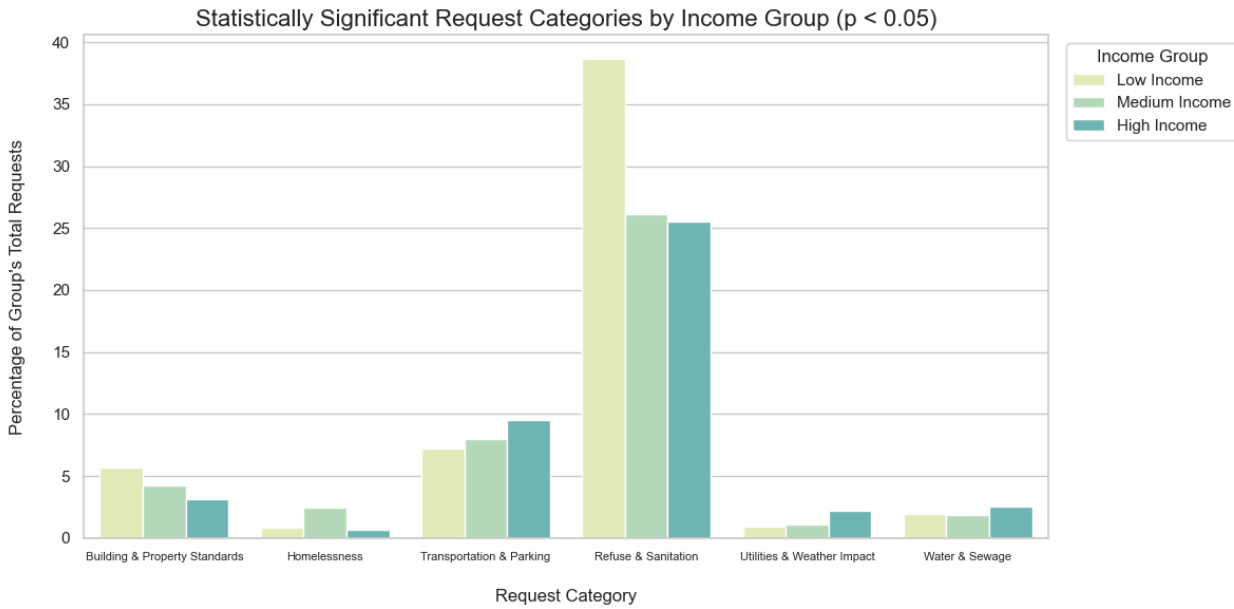
Table 7a: Kruskall-Wallis Test Results ($\alpha = 0.05$) (sample)

Request Category	Kruskal-Wallis Statistic (χ^2)	P-Value	Conclusion
Noise & Nuisance	4.81	0.0902	No significant relationship
Street & Infrastructure	0.51	0.7754	No significant relationship
Transportation & Parking	6.90	0.0317	**Significant relationship**
Water & Sewage	6.06	0.0484	**Significant relationship**

Refuse & Sanitation	14.86	0.0006	**Significant relationship**
Homelessness	7.93	0.0190	**Significant relationship**

Because of these interesting findings, another grouped bar chart (shown in figure 3) was created to investigate further:

Figure 3: Grouped Bar Chart of Statistically Significant Request Categories by Income Group (p < 0.05)



Upon further analysis, it appears that for these statistically significant categories, there isn't all that much variation by income group. The only exception here is low income reporting Refuse and Sanitation, but this seems to be an outlier based on the rest of the data not showing much of a pattern of income group influencing request type.

As opposed to the findings from Wang et al.'s 2017 study, this research finds that 311 data patterns generally cannot be used to create distinct socioeconomic signatures in cities like Pittsburgh. Median income is not a very reliable predictor of how often residents will submit certain types of requests, even for request categories found to show statistically significant relationships using Kruskall-Wallis testing.

4. Discussion

4.1 Substantive Findings

The research question asked if 311 service request patterns could predict socioeconomic status (specifically median income) in a medium-sized city like Pittsburgh in the same way Wang et al. found it could for huge cities like New York, Chicago, and Boston. The data and analysis reveal that 311 data is an inconsistent predictor in this way. While Wang et al. found that 311 complaint signatures could predict socioeconomic characteristics in huge cities, the data in Pittsburgh shows that there isn't much difference in the frequency of any given request category based on socioeconomic status.

4.2 Methodological Insights

Scaling down the methodology of the Wang research required some changes to the methodology initially unexpected in the research. The original 247 complaint types in the WPRDC 311 data were too noisy to yield helpful results, so some data aggregation had to be done–they were grouped into 24 request categories for analysis. On the other hand, the Wang et al. study was able to keep the 311 complaint data as-is. The predictive machine learning algorithms used by Wang et al. in their research were too complex and out of scope for this research, but the modified methods of creating income groups, only using median income, and using statistical testing worked to understand the general usefulness of 311 data as predictors of socioeconomic status. The Kruskall-Wallis test used was the correct choice – it avoided false positives a Chi-square test would have triggered due to skewed population numbers in the low income group, and instead focused on testing for variance between request frequencies for independent ordinal groups.

4.3 Limitations

One large limitation of using 311 data to predict socioeconomic status or characteristics is that it doesn't reflect the actual frequency of issues in a city, but the amount the population reports them. 311 data also inherently centers on the voices of "active" citizens who possess the appropriate resources (internet access and mobile devices); knowledge of the 311 system; trust in government institutions; and motivation that is required for someone to submit a 311 service request. Also, grouping the complaint types into broader categories – while helpful for analysis – means there was a loss in granularity, along with the income groupings as well. The methodological switch of not using machine learning models like the original research means that this isn't a perfect 1:1 recreation of the research. However, for the purposes of this paper, it absolutely gives a good enough approximation of how 311 data might or might not be helpful in a city like Pittsburgh.

5. Ethics

5.1 Ethical Considerations in this research

When planning this research, multiple ethical tensions were evaluated using Salganik's four ethical principles outlined in *Bit by Bit: Social Research in the Digital Age*.[5] In terms of

[5] Matthew J. Salganik, "Ethics" in *Bit by Bit: Social Research in the Digital Age* (Princeton University Press, 2018), Chapter 6.

respect for persons, while the 311 data being used is de-identified and public, the risk of reidentification exists if any extremely granular location data is linked with census data. Another concern is with the butterfly effects of relying on 311 data. 311 data is not a perfect reflection of the thoughts of all citizens. It's not even necessarily representative of the thoughts of most or many citizens. While this data might be helpful for planners or researchers, this data might be skewed towards certain demographics and what they want done in the city. If 311 data is able to accurately predict demographic data, this might result in unfair categorization of neighborhoods and might play into discriminatory practices by development entities or city planners.

5.2 How they're addressed

To uphold respect for law and public interest, this research only uses publicly available data on open-data portals and public US government sites. To mitigate privacy / re-identification risks, any unnecessary data not relevant to the research – longitude/latitude, timestamps, exact addresses, etc. – was discarded first and foremost. The data cleaning portion of this research was taken seriously to ensure that extremely granular data wasn't being analyzed when it wasn't necessary for the research. The data from these sites is also used according to each site's terms of service.

5.3 Reflection on ethical tradeoffs in this research

There are definite ethical tradeoffs in this research. On one hand, beneficence is high in this research. 311 is a free data source and tool that can be used for urban planning and by researchers, and therefore has the opportunity to benefit many citizens for relatively low cost. This data source is massive, open, free, and granular, and can be used by researchers like Wang et al. to understand cities much cheaper and quicker than a census could. As alluded to above, however, justice is a point of tension, though. If 311 data is used to decide where to allocate resources, neighborhoods that don't have access to tools needed to submit 311 requests (or that don't know about or know how to use these tools) will be less represented in this data. In other words, already-marginalized groups might be made more marginalized in some aspects if relying on this data. The ethical tradeoff here is that by using highly available, granular, public data might reinforce an inherently disproportionate system that misses the needs of the marginalized voices.

6. Conclusion

This paper investigated whether 311 service requests could create reliable signatures for neighborhood socioeconomic status in Pittsburgh like Wang et al. found they can in huge cities like New York City, Chicago, and Boston. After categorizing over 25,000 requests into 24 thematic categories and applying Kruskall-Wallis testing, the results revealed that request category is not a reliable predictor of median income (the stand-in for the multiple socioeconomic characteristics studied by Wang et al.). Even for the 6/24 categories where a statistically significant relationship was found, further investigation revealed little variation between income groups. In conclusion, 311 data cannot be used to create complaint signatures in cities like Pittsburgh in the same way it can in New York City and other huge cities as found by Wang et al.

The primary contribution of this research is the demonstration that the power of 311 data is very dependent on the scale of the city. While Wang et al. established that 311 could accurately predict demographics in high-density cities like New York, Boston, and Chicago, this research's method swap highlights the limitation of 311 data use in smaller cities like Pittsburgh with a population around 300,000. This research may also help future researchers looking to utilize 311 data in smaller urban contexts. The methods used in this research offer a way to extract meaningful insights from 311 data without the scale or volume provided by a huge city like New York City.

All in all, this paper serves as a reminder to urban stakeholders and researchers that while 311 data is a valuable civic tool and a potentially helpful aid for understanding an area, it should be approached and used with caution, especially when it comes to predicting the demographics of neighborhoods in smaller urban areas.

7. References

Bibliography

Lemery, Christopher. "Guides: Pittsburgh Census Information @ Pitt: Pittsburgh Census Tracts, 1940-2020." Accessed April 22, 2026. https://pitt.libguides.com/pghcensus/pghcensustracts.

Salganik, Matthew J. Bit by Bit: Social Research in the Digital Age. Princeton University Press, 2018. Accessed April 23, 2026. https://www.bitbybitbook.com/en/ethics/

United States Census Bereau. "City and Town Population Totals: 2020-2024." Census.Gov. Accessed April 22, 2026. https://www.census.gov/data/tables/time-series/demo/popest/2020s-total-cities-and-towns.html.

United States Census Bereau. "DP03 - Census Bureau Tables." Accessed April 22, 2026. https://data.census.gov/table?q=DP03&g=160XX00US4261000,4261000$1400000&d=ACS+5-Year+Estimates+Data+Profiles.

Wang, Lingjing, Cheng Qian, Philipp Kats, Constantine Kontokosta, and Stanislav Sobolevsky. "Structure of 311 Service Requests as a Signature of Urban Location." PLOS ONE 12, no. 10 (2017): e0186314. https://doi.org/10.1371/journal.pone.0186314.

Western Pennsylvania Regional Data Center. "311 Data Archive - 311 Data - CKAN." Accessed April 22, 2026. https://data.wprdc.org/dataset/311-data/resource/29462525-62a6-45bf-9b5e-ad2e1c06348d.

Western Pennsylvania Regional Data Center. "2020 Census Redistricting Data Extracts (PL 94-171) - Block Index City of Pittsburgh Neighborhoods - CKAN." Accessed April 22, 2026. https://data.wprdc.org/dataset/2020-census-redistricting-data-extracts/resource/6b09ea3e-7d34-4665-ad0b-798a0efadc29.

8. Appendix

8.1 Code

Complete code for this paper can be found in this Jupyter Notebook:
https://drive.google.com/file/d/1OBrEpqLXRiQbEaRTOfs7ARMXYF5kkMK_/view?usp=drive_link
This notebook also contains full data tables, some of which aren't referenced in the paper but were generated with code.

And in this GitHub repo: https://github.com/npisar/SI313-homework/tree/main/final%20project

8.2 AI Appendix

- I used GenAI throughout the process of writing this paper and the creation of the code. I fed AI progress report PDFs and Jupyter notebook files to help create outlines of the main sections of this paper, giving me a helpful starting point to get the bones of the paper complete. For example, in my initial prompt to create the outline, I said:
    - "Based on this notebook and pdf so far, please draft an outline that hits on the following key points: [copy pasted assignment details]"
- I also used AI to help me expand out sections like my methodology. These were things I had already done in my code and progress report assignments, so this was helpful to speed things up. Here is an example prompt:
    - "Please expand on how my methods differ from the Wang et al paper"
- I used AI as a formatting tool to save time on things like data entry and table conversion from my code to the paper. I pasted raw python text output to AI and asked it to create data tables I could copy and paste into my report:
    - "I need to take this mapping and create a table that I can use for my final paper. Please create a table that shows each of the main categories, and what complaint types were mapped to each individual category. It should be ready to copy and paste into a google docs table"
    - "Can you turn this Kruskal Wallis test data into a table that's ready to copy paste into google docs?" (After pasting in the raw text output of my 24 statistical tests).
- Like in previous projects, I used AI to help with coding and graphing. I figured out I needed to change my analytical statistical approach from Chi-square to Kruskall-Wallis midway through the research, so I got help understanding how to use that test in python. I had never used it before.
- I used AI to help generate the heatmap and grouped bar chart code; it's much faster than me and dynamically adapts to my data structures when I append my notebook files.
- I used AI to create the thematic mapping of request categories. This was much faster than doing it by hand and helped make the analysis actually practical.
- I didn't know what "melting" a dataframe was in some of the sample code that I got back from AI when I asked it to help create visualizations. I asked it to explain what this function does so I could effectively work with the data.

- **In general, I used AI on this assignment like I have for previous assignments.** I generated outlines, starting points for code, and got the building blocks done with AI, but I used my own analysis and logic to actually flesh it out. For example, AI originally suggested I used a chi-square test, but I realized it was wrong for this research and changed to the Kruskall-Wallis test instead.

8.3 Tables / Figures

Table 1: Complaint Type Mapping

Category	Mapped Complaint Types
Refuse & Sanitation	Missed Refuse Pick Up, Missed Recycling Pick Up, Bulk Pick Up, Special Pick Up, Courtesy Pick Up, Illegal Dumping, Dumping, Private Property, Early Set Out, Late Set Out, Litter, Litter, Public Property, Weeds/Debris, Recycling Violation, Refuse Violations, Blue Bin Containers, Dumpster, Dumpster (on Street), Commercial Refuse/Dumpsters, Litter Can, Litter Can, Public, Dead Animal, Animal Waste, Rodent control
Street & Infrastructure	Potholes, Paving Concern/Problem, Paving Request, Brick or Block Repair, Concrete Slab Repair, Street Cleaning/Sweeping, Landslide, Sinkhole, Bridge Maintenance, Curb Cuts, Curb/Request for Asphalt Windrow, Manhole Cover, Manhole Covers, PWSA, Utility Cut - Other, Utility Cut - PWSA, City Cuts Concern, Sidewalk/Curb/ADA Ramp Maintenance, Broken Sidewalk, Repair City Steps
Transportation & Parking	Illegal Parking, Abandoned Vehicle (parked on street), Towed Vehicle, Junk Vehicles (Private Property), Speeding, Traffic Enforcement, Traffic Calming, Loading Zone, Request, Permit Parking (Request for RPP zone), Parking Authority, Ticket, Pgh Parking Authority, Ticket, Police-issued, SPIN (Stand Up) Scooters, Boat/Trailer on Street
Water & Sewage	Catch Basin, Clogged, Collapsed Catch Basin, Water Main Break, Leak, Drainage/Leak, Flooding, Sewage, Sewers, Stormwater Runoff, Water Pressure, No Water
Building & Property Standards	Building Maintenance, Building Without a Permit, Vacant Building, Vacant Lot, Board Up (City-owned property only), Fence Maintenance, Zoning Issue, Overgrowth, Couch on Porch, Unpermitted Electrical Work, Unpermitted Fire System Work
Noise & Nuisance	Noise, Excessive Noise/Disturbances, Nuisance Bar, Barking Dog, Panhandling
Parks & Greenery	Pruning (city tree), Tree Removal, Tree Issues, Stump Grind/Removal, Root prune, Dead tree (Public property), Tree Fallen Across Road, Playground, Ballfield, Park Shelter, Overgrowth, Parks, Dead Tree (3TB), Tree Fallen Across Sidewalk
Safety & Lighting	Street Light - Repair, Street Light - Request, Lights, Drug Enforcement, Gang Activity, Police - Submit a Tip, Fire Safety System Issue, Wires
Snow & Ice Removal	Snow/Ice removal, Sidewalk, Lack of Snow/Ice Removal, Sidewalk has Ice or Litter, City Steps, Need Cleared, Salt Box, Snow Angel Concern
Traffic Signals & Signs	Replace/Repair a Sign, Traffic or Pedestrian Signal, Repair, Request New Sign, Traffic Shop, Traffic Signal Request, Traffic or Pedestrian Signal, Request, Pedestrian Signal Maintenance, Signs, Advertising or Political, Sign, Traffic Signals Surtrac, Pedestrian Signal Request
Active Transportation	Bicycle/Pedestrian Concerns, Bike Lane Bollard, Blocked or Closed Sidewalks, Blocked or Closed Trails, Trail Maintenance, Bicycle Parking, Crosswalk and Street Markings, Maintenance, Crosswalk, New, Pavement Marking, New

Category	Mapped Complaint Types
Traffic Safety & Crashes	Dashcam, Crashes and Close Calls, Traffic Improvement, Barricades, Crossing Guards
Right of Way & Obstructions	Street Obstruction/Closure, Road, Public Right of Way, ROW, Construction Site Maintenance, Guide Rail, Angle Iron
City Property Maintenance	City Owned Property Maintenance, City Facility, Water/Drinking Fountains, Bus Shelter, Bus Stop Request, PRT, County Maintenance, URA property, Land Bank, Shelters, Port A Potty
Accessibility (ADA)	Americans with Disabilities, Accessible Parking Signs, Removal, Business Accessibility, Accessibility Construction Issue, Accessible Parking Application, ADA Ramp, Installation
Graffiti & Vandalism	Graffiti in Right of Way, Graffiti on Structure
Homelessness	Homeless
Govt Accountability	Misconduct/City Employee, 911 Performance, Ethics Office, Commission on Human Relations, MO Escalated
Property & Business Compliance	Operating Without a License, Tenant/Landlord Problems, Electrical Violation, HVAC Not Functioning, Plastic Bag Ban Compliance, Unpermitted Land Operations, Building Benchmark Ordinance, Improper Work in a Historic District, Food Safety Complaints, Paid Sick Leave, Unpermitted HVAC Work, Unpermitted Sign Construction
Tax & Financial Admin	Real Estate Tax, Earned Income Tax, Homestead Act/Senior Tax Relief, EMS Billing, PWSA Billing or Shut Off
Animals & Wildlife	Loose Dog(s), Deer Management Pilot Program, Dog License, Off Leash Exercise Area
Parks & Rec (Facilities)	Field, Parks Trails, Court, Basketball or Tennis, Park Ranger, Greenways, Basketball Hoop
Utilities & Weather Impact	Storm Damage, Hydrant, Utility Pole, Leaves/Street Cleaning, Leaves, Grass or Other Yard Debris

Table 2: Top 5 Most Common Request Categories in Pittsburgh

Request Category	Total Count
Refuse & Sanitation	8144
Street & Infrastructure	2863
Transportation & Parking	1998
Safety & Lighting	1633
Traffic Signals & Signs	1245

Table 3a: Top 5 request categories per neighborhood (sample)

Full 382 row table available in code / notebook linked above

neighborhood	request_category	relative_frequency	percentage
Allegheny Center	Snow & Ice Removal	0.184834	18.48
Allegheny Center	Safety & Lighting	0.174171	17.42
Allegheny Center	Traffic Safety & Crashes	0.150474	15.05
Allegheny Center	Street & Infrastructure	0.116114	11.61
Allegheny Center	Active Transportation	0.071090	7.11

Table 4: Descriptive statistics - median income

measure	value
count	25153.000000
mean	72420.996303
std	36713.688650
min	8889.000000
25%	52099.000000
50%	68708.000000
75%	83500.000000
max	222778.000000

Table 5: Top 5 request categories per income group

income_group	request_category	relative_freq
Low Income	Refuse & Sanitation	0.386511
Low Income	Street & Infrastructure	0.102300
Low Income	Transportation & Parking	0.071969
Low Income	Safety & Lighting	0.065653
Low Income	Building & Property Standards	0.056686
Medium Income	Refuse & Sanitation	0.261577
Medium Income	Street & Infrastructure	0.110763
Medium Income	Transportation & Parking	0.079631
Medium Income	Safety & Lighting	0.063830
Medium Income	Traffic Signals & Signs	0.058041
High Income	Refuse & Sanitation	0.255222
High Income	Street & Infrastructure	0.142015
High Income	Transportation & Parking	0.095350
High Income	Safety & Lighting	0.064522
High Income	Parks & Greenery	0.061152

Table 6: Request Categories with Highest Standard Deviation

request_category	std dev
Refuse & Sanitation	7.4
Street & Infrastructure	2.09
Parks & Greenery	1.68
Active Transportation	1.49
Snow & Ice Removal	1.46

Table 7: Kruskall-Wallis Test Results ($\alpha = 0.05$)

Request Category	Kruskal-Wallis Statistic (χ^2)	P-Value	Conclusion
Accessibility (ADA)	0.16	0.9238	No significant relationship
Active Transportation	4.31	0.1159	No significant relationship
Animals & Wildlife	1.07	0.5865	No significant relationship
Building & Property Standards	7.60	0.0224	**Significant relationship**
City Property Maintenance	0.06	0.9695	No significant relationship
Govt Accountability	0.11	0.9469	No significant relationship
Graffiti & Vandalism	0.04	0.9817	No significant relationship
Homelessness	7.93	0.0190	**Significant relationship**
Noise & Nuisance	4.81	0.0902	No significant relationship
Parks & Greenery	0.44	0.8028	No significant relationship
Parks & Rec (Facilities)	0.94	0.6255	No significant relationship
Property & Business Compliance	3.27	0.1945	No significant relationship
Refuse & Sanitation	14.86	0.0006	**Significant relationship**
Right of Way & Obstructions	4.35	0.1137	No significant relationship
Safety & Lighting	0.79	0.6740	No significant relationship
Snow & Ice Removal	1.84	0.3988	No significant relationship
Street & Infrastructure	0.51	0.7754	No significant relationship
Tax & Financial Admin	4.54	0.1032	No significant relationship
Traffic Safety & Crashes	4.13	0.1268	No significant relationship
Traffic Signals & Signs	2.57	0.2769	No significant relationship
Transportation & Parking	6.90	0.0317	**Significant relationship**
Uncategorized/Other	0.51	0.7767	No significant relationship
Utilities & Weather Impact	10.99	0.0041	**Significant relationship**
Water & Sewage	6.06	0.0484	**Significant relationship**

Table 7a: Kruskall-Wallis Test Results ($\alpha = 0.05$) (sample)

Request Category	**Kruskal-Wallis Statistic (χ^2)**	**P-Value**	**Conclusion**
Noise & Nuisance	4.81	0.0902	No significant relationship
Street & Infrastructure	0.51	0.7754	No significant relationship
Transportation & Parking	6.90	0.0317	**Significant relationship**
Water & Sewage	6.06	0.0484	**Significant relationship**
Refuse & Sanitation	14.86	0.0006	**Significant relationship**
Homelessness	7.93	0.0190	**Significant relationship**

Figure 1: Heatmap of Relative Frequencies of Request Categories by Income Group

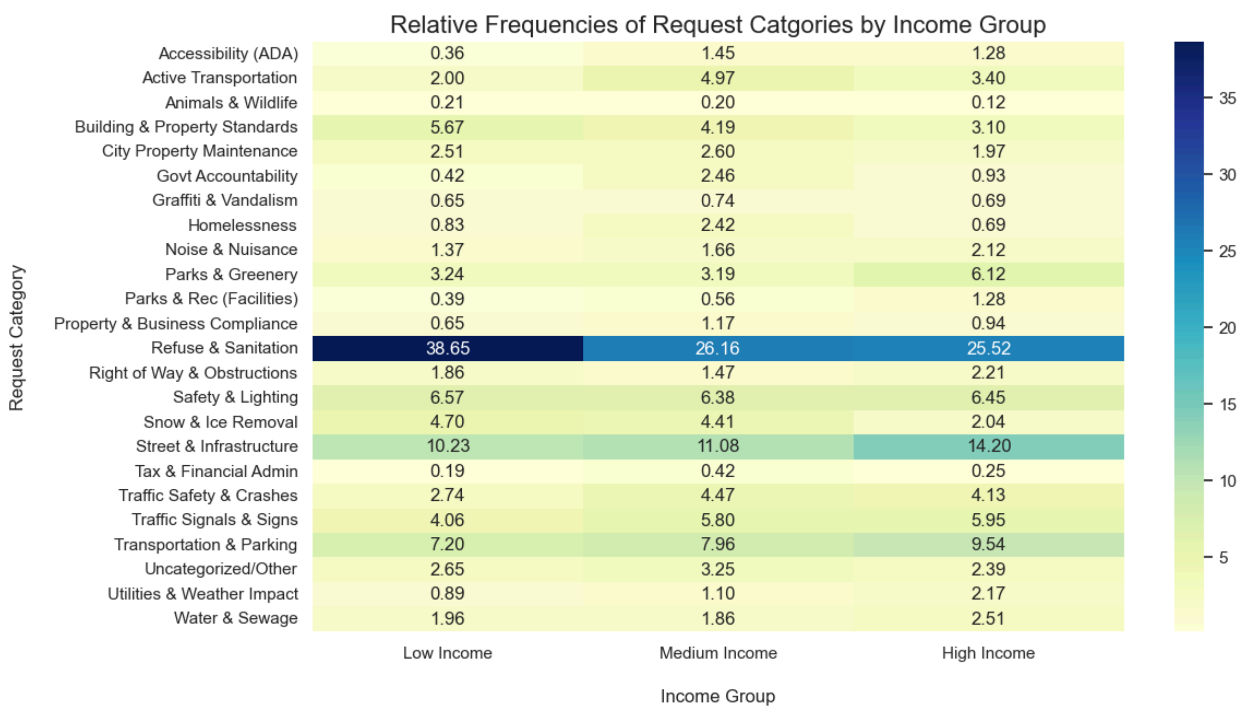


Figure 2: Grouped Bar Chart of Top 5 Request Categories with Highest Standard Deviation by Income Group

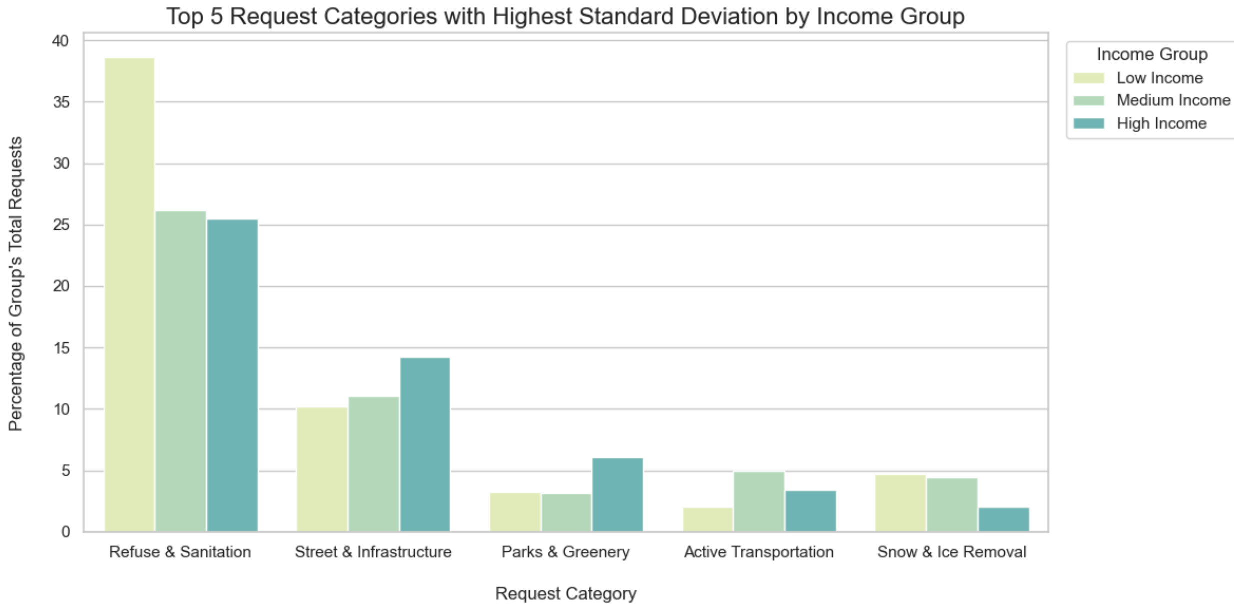


Figure 3: Grouped Bar Chart of Statistically Significant Request Categories by Income Group (p < 0.05)